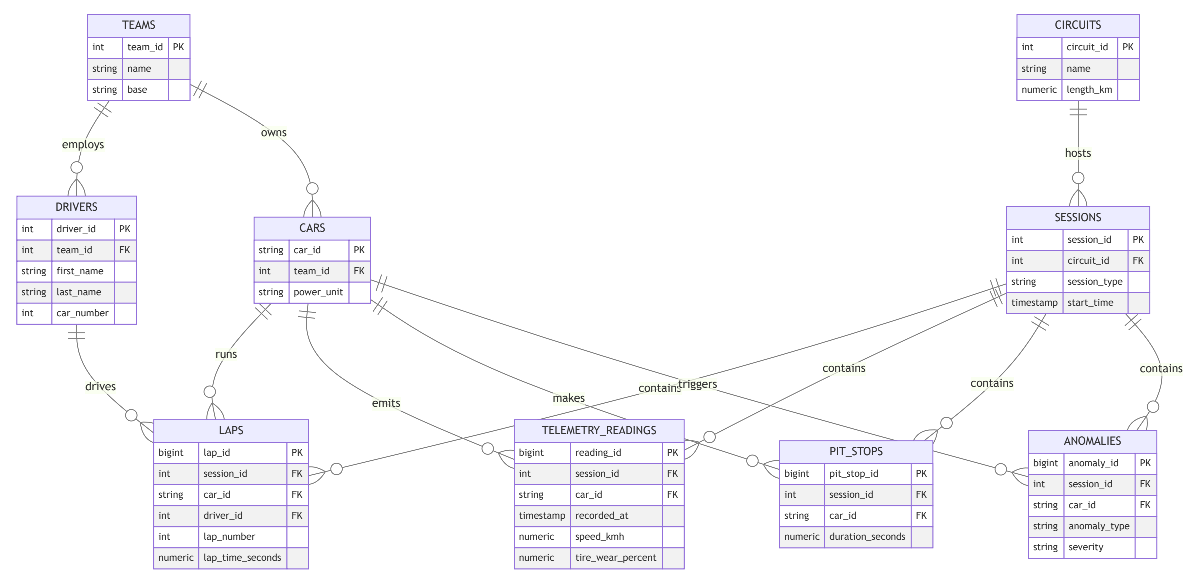
Data Model

Two layers, both PostgreSQL (DDL in [`../sql/ddl`](../sql/ddl)):

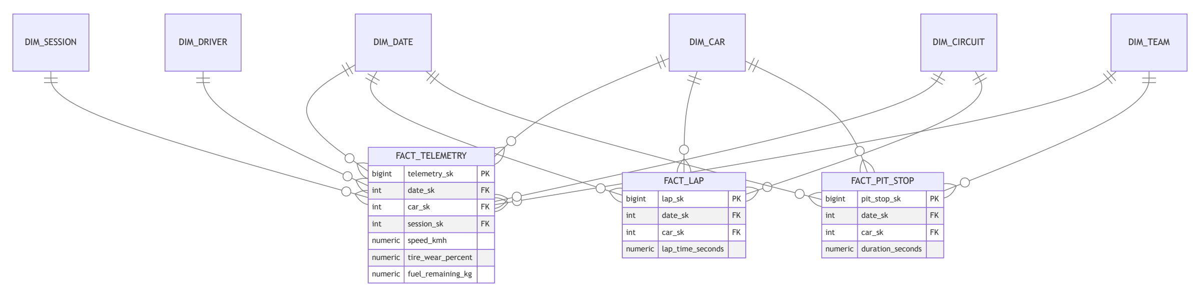
1. **Operational (3NF)** - the source of truth for historical telemetry.
2. **Analytics (star schema)** - dimensional model built from the operational tables for fast aggregation and BI/Grafana.

1. Operational model (ERD)



2. Analytics model (star schema)

Conformed dimensions shared by all facts; surrogate keys (`_sk`) decouple the warehouse from operational IDs and leave room for slowly-changing dimensions.



3. Design rationale

Decision	Why
Split OLTP (3NF) and OLAP (star)	Normalised tables keep writes consistent and storage lean; the star schema makes analytical aggregations (avg lap time, tire wear over a stint, pit-stop duration by team) fast and BI-friendly.
Grain of `fact_telemetry` = one reading/car/timestamp	Finest useful grain; everything else (per-lap, per-stint, per-session) rolls up from it.
Surrogate keys on dimensions	Decouple analytics from source IDs; enable SCD Type 2 (e.g. a driver changing teams) without breaking facts.
`dim_date` (YYYYMMDD)	Standard date dimension for time-series rollups and Grafana time filters.
Index `telemetry_readings (session_id, car_id, recorded_at)` and `fact_telemetry (date_sk, car_sk, session_sk)`	The dominant query pattern is "a car's telemetry over time in a session".
PostgreSQL (RDS)	Already used by Airflow; managed, encrypted, point-in-time recovery. For multi-month, high-cardinality telemetry, TimescaleDB / a columnar warehouse (BigQuery, ClickHouse) is the next step.

The real-time path (Prometheus + Grafana) covers short windows; this relational model is the durable historical/analytical store, loaded by the Airflow batch DAG.